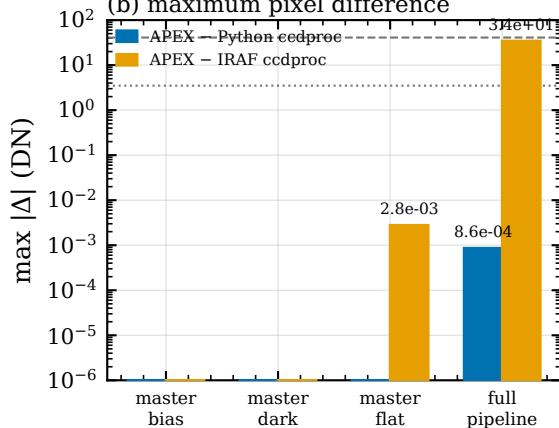


(a) pixel-level comparison

stage	Python max $ \Delta / \sigma$	IRAF max $ \Delta / \sigma$	interpretation
master bias	0 0	0 0	bit exact
master dark	0 0	0 0	bit exact
master flat	0 0	2.8e-03 1.5e-03	flat norm.
full pipeline	8.6e-04 2.6e-05	3.4e+01 9.0e-01	chain arithmetic

entries are max |APEX – reference| / robust σ ; 0 means every pixel matched
IRAF flat was divided by its median before comparison (same convention as APEX)

(b) maximum pixel difference



NGC 6811 B 60 s · Moravian C3-61000 (2×2) · 2026-06-11 · 8 bias / 8 dark (60 s) / 5 flat

same raw frames and median-combine convention · IRAF ccdred/PyRAF · cosmetic repair disabled